

Negative-Weight Reweighting

The algebra, and why it is an identity

$H \rightarrow WW + \text{charm} \cdot V + \text{jets} \cdot \text{arXiv:2510.16217}$

1 · Where negative weights come from

An NLO cross section splits into real and virtual contributions:

$$\sigma_{\text{NLO}} = \int d\Phi_{n+1} R + \int d\Phi_n V + \int d\Phi_n B$$

R and V are **each divergent**; the divergences cancel only in the sum. To integrate numerically you introduce subtraction terms S :

$$\sigma_{\text{NLO}} = \int d\Phi_{n+1} (R - S) + \int d\Phi_n \left(V + \int_1 S \right) + \int d\Phi_n B$$

Each bracket is now finite — but $R - S$ and $V + \int_1 S$ are **not positive definite**. Events sampled from them carry weights of either sign.

Negative weights are not a bug or a generator pathology. They are the price of a finite, matched NLO calculation.

2 · Why that hurts

The yield estimator in a bin B :

$$\hat{N}_B = \sum_{i \in B} w_i, \quad w_i = \pm |w_i|$$

quantity	behaviour
expectation $\langle \hat{N}_B \rangle$	correct — the estimator is unbiased
variance $V[\hat{N}_B]$	$\propto \sum_i w_i^2 = \sum_i w_i ^2$

The **mean** is the signed sum (small), the **variance** follows the unsigned sum (large). Define the effective sample size

$$n_{\text{eff}} = \frac{\left(\sum_i w_i\right)^2}{\sum_i w_i^2} \ll N$$

In our V+jets, **16.4%** of events had $w < 0$. One DY bin came out 0 ± 41 from $\pm 79,000$ weights cancelling — zero content, large error. `autoMCStats` then assigns that bin a huge nuisance.

3 · The decomposition

Any signed density can be written as a **difference of two positive, normalised** densities:

$$p(\vec{x}) = a p_+(\vec{x}) - b p_-(\vec{x})$$

with $a, b \geq 0$ and, since p is itself normalised,

$$\int p = a - b = 1$$

p_+ — the normalised density of **positive-weight** events

p_- — the normalised density of **negative-weight** events

a, b — their total weights, with $a - b = 1$

A Monte Carlo generator samples p_+ with probability $\propto a$ and p_- with probability $\propto b$, assigning $\pm$ accordingly.

4 · The positive-weight probability

At a point $\vec{x}$, the probability that an event drawn there carries $w > 0$:

$$P_+(\vec{x}) = \frac{a p_+(\vec{x})}{a p_+(\vec{x}) + b p_-(\vec{x})}$$

Write $u(\vec{x}) \equiv a p_+(\vec{x}) + b p_-(\vec{x})$ — the **unsigned** density.

Then

$$a p_+ = P_+ u, \quad b p_- = (1 - P_+) u$$

Substituting into $p = a p_+ - b p_-$:

$$p(\vec{x}) = P_+ u - (1 - P_+) u = (2P_+(\vec{x}) - 1) u(\vec{x})$$

$$p(\vec{x}) = g(\vec{x}) u(\vec{x}), \quad g(\vec{x}) \equiv 2P_+(\vec{x}) - 1$$

5 · Why this solves the problem

$u(\vec{x})$ is the density that $\sum_i |w_i|$ samples — **all events, unsigned**.

So for any observable f :

$$\begin{aligned} \int f(\vec{x}) p(\vec{x}) d\vec{x} &= \int f(\vec{x}) g(\vec{x}) u(\vec{x}) d\vec{x} \\ \implies \sum_i w_i f(\vec{x}_i) &\equiv \sum_i |w_i| g(\vec{x}_i) f(\vec{x}_i) \end{aligned}$$

Both estimate the same distribution. The reweighting is therefore

$$w_i \longrightarrow |w_i| \cdot g(\vec{x}_i)$$

Same yield. Same shape. Smaller variance — because every event now contributes with the same sign, scaled by $g \in [-1, 1]$, instead of cancelling.

6 · The variance gain, explicitly

	signed	reweighted
estimator	$\sum w_i$	$\sum w_i g_i$
per-event magnitude	$ w_i $	$ w_i \cdot g_i \leq w_i $
cancellation	yes	no

Since $|g| \leq 1$, every term shrinks *and* nothing cancels. The variance

$$V\left[\sum |w_i| g_i\right] = \sum |w_i|^2 g_i^2 \leq \sum w_i^2$$

with equality only where $|g| = 1$, i.e. where the sign is deterministic.

Measured in our sample: $\langle g \rangle = 0.672 = 2(0.836) - 1$, matching the global positive fraction. Range $[-0.991, 0.993]$ — inside $[-1, 1]$, no pathological events.

7 · The one approximation

Everything above is **exact for the true P_+** . In practice P_+ is unknown, so we estimate it with a classifier:

$$\hat{P}_+(\vec{x}) \approx P_+(\vec{x}) \implies \hat{g} = 2\hat{P}_+ - 1$$

This is the only approximation in the method. Closure stops being an algebraic guarantee and becomes an **empirical property that must be measured**.

model	HistGradientBoostingClassifier, 20-model ensemble
target	$\text{sign}(w)$
features	20, all generator-level (<code>lhe_*</code> , <code>genparton_*</code>)
training	9.8M events
ensemble AUC	0.829

8 · Why generator-level features only

$P_+(\vec{x})$ is a property of **the generator**, not of the detector.

The identity $p = g u$ holds in the variables the generator actually sampled. If $\vec{x}$ contained reconstruction-level quantities, u would no longer be the density $\sum |w|$ samples, and closure would not follow.

The paper (§V.2) rejects reco-level inputs for exactly this reason: *"using reconstruction-level variables will not guarantee closure on generator-level variables."*

Our 20 features: `lhe_njets`, `lhe_nb/nc/nuds/nglu`, `lhe_ht`, `lhe_vpt`, `lhe_alphas`, `genparton_multiplicity`, `genparton_n_pt20/100/200`, incoming parton PDG IDs, and the two leading partons' p_T and η .

9 · Train loose, infer tight

The signal region has far too few V+jets events to train on — that starvation is the problem being solved.

	region
train	all base cuts + veto of the $e\mu$ SR topology
infer	the tight $e\mu$ signal region

Disjoint by construction. An event that is not "exactly one $e\mu$ pair" can never be an SR event, so training and inference sets cannot overlap — no event-ID bookkeeping needed, and no overfitting bias.

Disjoint **events**, but *not* an orthogonal **phase space**. g must be interpolated over the SR's support, never extrapolated — so an anti-SR training region (e.g. c-depleted) would be actively harmful.

10 · Validation

check	result
training-region closure	0.994 on 9.8M events
calibration of $\hat{P}_+$ in the SR	on the diagonal, $P_+ \approx 0.15 \rightarrow 0.93$
ensemble spread δg	mean 0.006 , max 0.467
SR closure	carries an offset

The SR closure offset is real. The SR is a small, kinematically-biased corner of the training space, so a finite $\hat{P}_+$ does not close perfectly there.
This caveat is our extension — the paper's evaluation region was not this starved.

Why it is acceptable: reweighting fixes **variance, not yield**; the n_{eff} benefit comes from reduced per-bin variance and survives the offset; and the residual is covered by `CMS_negrw_vjets`, which **profiles to 0.0%** in the fit.

11 · What enters the analysis

```
# per event, dumped alongside the nominal weight
weight_negrw      = 2 * P_plus_ensemble_mean(x) - 1      # = g
weight_negrw_std  = 2 * P_plus_ensemble_std(x)           # = delta g
```

column	role
weight_negrw	the reweight factor g applied to $ w $
weight_negrw_std	the 20-model spread → shape nuisance CMS_negrw_vjets

Applied only to the V+jets samples the ensemble was trained for, via an **anchored** dataset match — a substring gate would also catch `Wp1usH_WtoLNu...` and silently reweight a Higgs template with a V+jets model.

`CMS_negrw_vjets` profiles to **0.0%** — the method's own uncertainty is negligible.

Summary

$$p(\vec{x}) = (2P_+(\vec{x}) - 1) u(\vec{x})$$

An algebraic identity, not an approximation.

Same yield, same shape, no cancellation.

The only approximation is $\hat{P}_+$ from a finite classifier —
and its residual is a nuisance that profiles to zero.